

Introduction to Robotics, Lecture 2

Direct kinematics and DH notation

Analytical inverse kinematics

Numerical methods and workspaces

MIPT

Center for Cognitive Modeling

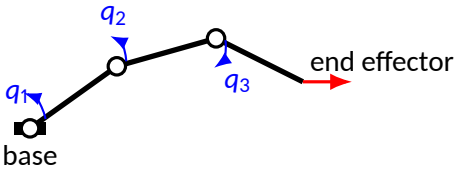
What Kinematics Studies

Motion geometry before forces and torques

Kinematics

The study of the **geometric** and **timing** aspects of robot motion, without modelling the causes that produce it.

A manipulator is modelled as an open chain of **rigid links** connected by revolute or prismatic joints.



Geometry answers *where*. Dynamics later explains *why the motion changes*.

Why Compute Kinematics?

The geometric model connects a task to executable motion

Understand the robot

- determine the reachable workspace
- calibrate the geometric model
- predict the end-effector pose

Make the robot move

- define tasks in Cartesian space
- plan and program trajectories
- provide references to motion control

task description ⟶ kinematic model ⟶ joint motion

Two Descriptions of One Configuration

Joint space describes the mechanism. Task space describes what matters

Joint coordinates

$$q = \begin{bmatrix} q_1 & \cdots & q_n \end{bmatrix}^T \in \mathbb{R}^n$$

One coordinate per independent joint of an open chain.

Examples include a revolute angle and a prismatic displacement.

Task variables

$$r = \begin{bmatrix} r_1 & \cdots & r_m \end{bmatrix}^T \in \mathbb{R}^m$$

The selected position and orientation quantities needed by the task.

The same robot can use a different r for a different task.

Direct and Inverse Kinematic Maps

The forward map supports two different calculations

Direct kinematics

Given q, evaluate one task vector

$$r = f_r(q).$$

This calculation follows the chain from the base to the tool.

Inverse kinematics

Given r_d, solve the equation

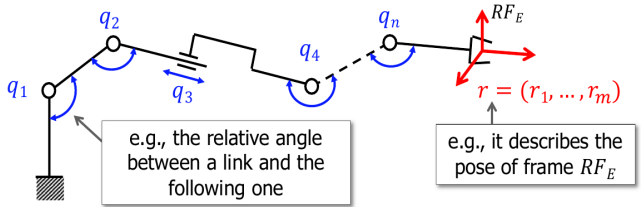
$$f_r(q) = r_d.$$

The solution set can contain zero, one, or several joint vectors.

The notation f_r^{-1} is unsafe when the inverse is not single valued.

Anatomy of an Open Kinematic Chain

Each downstream pose depends on every upstream joint



- number joints from the base
- attach local frames along the chain
- identify the terminal frame RF_E
- propagate motion from q_1 to q_n

$$q = [q_1, \dots, q_n]^T$$

Task Dimension Is a Modelling Choice

The task keeps the end-effector variables we intend to control

Task	Dimension m
planar point position	2
spatial point position	3
planar pose	3
spatial pose	6

Typical design goal

$$m \leq n$$

provides enough joint variables for an m-dimensional local task.

It does not guarantee global reachability, uniqueness, or freedom from singularities.

Two Routes to Direct Kinematics

Inspection is efficient. Frame propagation is systematic

Geometric inspection

- draw link vectors
- project them onto axes
- add relative angles

Homogeneous transforms

- attach a frame to every link
- describe each relative motion
- multiply in chain order

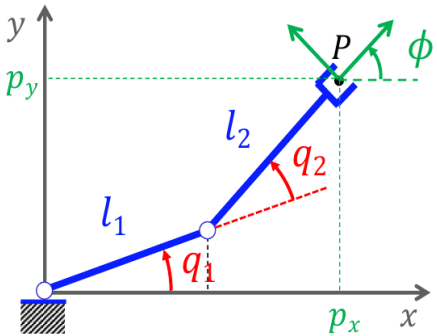
Best for simple planar mechanisms.

$$^{0}T_{n} = {}^{0}T_{1}{}^{1}T_{2}\cdots{}^{n-1}T_{n}$$

Both routes must produce the same end-effector pose.

Planar 2R Direct Kinematics by Inspection

Project each link and add the relative rotations



$$p_x = l_1 \cos q_1 + l_2 \cos(q_1 + q_2),$$
$$p_y = l_1 \sin q_1 + l_2 \sin(q_1 + q_2),$$
$$\phi = q_1 + q_2.$$

Dimension check

The robot has $n = 2$ joints but the pose vector $r = [p_x, p_y, \phi]^T$ has $m = 3$ entries. They cannot all be commanded independently.

Why a Frame Convention Is Needed

A fixed convention makes link transforms comparable

Any consistent set of link frames can describe a robot.

Arbitrary choices, however, produce parameter tables that are hard to compare and easy to misread.

Purpose of DH notation

Use the same four geometric parameters and the same motion order for every link.

$$^{i-1}T_i = T_z(d_i)R_z(\theta_i)T_x(a_i)R_x(\alpha_i)$$

Video checkpoint

DH frame assignment. Viewing prompt: which axis must coincide with joint *i*?

DH Notation

Four parameters describe one relative link transform

Parameter	Motion
θ_i	rotation about z_{i-1}
d_i	translation along z_{i-1}
a_i	translation along x_i
α_i	rotation about x_i

$$T_z(d_i)\,R_z(\theta_i)\,T_x(a_i)\,R_x(\alpha_i)$$

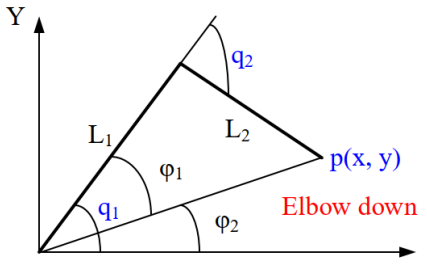
- revolute joint: θ_i varies
- prismatic joint: d_i varies

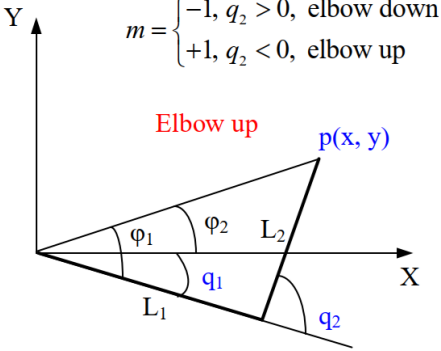
Convention check

Standard and modified DH use different orders. A table and its transform formula must use the same convention.

The 2R Arm as a Transform Chain

Forward kinematics assigns one pose to each joint pair





$${}^{0}T_1 = R_z(q_1)T_x(l_1), \qquad {}^{1}T_2 = R_z(q_2)T_x(l_2)$$

First Link Transform

The rotation block gives axes. The last column gives the elbow position

$$
{}^{0}T_{1} = \begin{bmatrix} c_1 & -s_1 & 0 & l_1 c_1 \\ s_1 & c_1 & 0 & l_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
c_i = \cos q_i, \qquad s_i = \sin q_i
$$

Read the matrix

The elbow coordinates are $(l_1 c_1, l_1 s_1)$.

Composition of Both Links

Matrix multiplication recovers the angle-sum terms

$$
{}^{0}T_{2} = {}^{0}T_{1}\,{}^{1}T_{2} = \begin{bmatrix} c_{12} & -s_{12} & 0 & l_1 c_1 + l_2 c_{12} \\ s_{12} & c_{12} & 0 & l_1 s_1 + l_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
c_{12} = \cos(q_1 + q_2), \qquad s_{12} = \sin(q_1 + q_2)
$$

Composition rule

Relative transforms multiply in physical chain order. Reversing the factors describes a different motion.

Geometry and Matrices Agree

The two derivations provide a direct implementation check

From vector projection

$$p_x = l_1 c_1 + l_2 c_{12},$$
$$p_y = l_1 s_1 + l_2 s_{12},$$
$$\phi = q_1 + q_2.$$

From ${}^{0}T_{2}$

- last column gives (p_x, p_y)
- rotation block gives ϕ

A mismatch usually means a frame, sign, relative-angle, or multiplication-order error.

The Inverse Kinematics Problem

A desired task value defines a nonlinear equation, not a guaranteed inverse

Given

A desired end-effector pose or task vector r_d.

Find

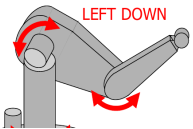
Every admissible joint vector q satisfying

$$f_r(q) = r_d.$$

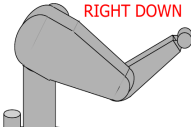
Existence	Multiplicity	Method
Is r_d reachable?	How many branches?	Closed form or iteration?

Configuration Branches of a 6R Arm

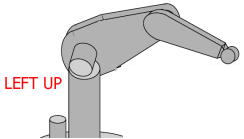
Shoulder and elbow choices already create four main-arm solutions



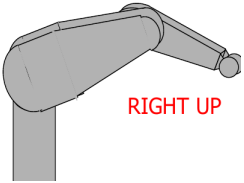
left / elbow down



right / elbow down



left / elbow up



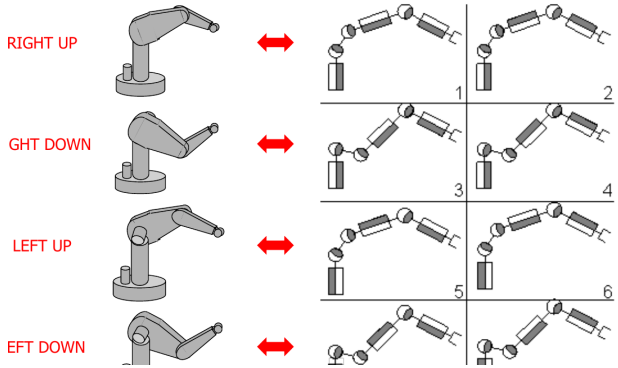
right / elbow up

Main-arm branch count

$2_{\text{shoulder}} \times 2_{\text{elbow}} = 4$ regular position branches away from singularities.

Why Four Position Branches Become Eight Poses

A spherical wrist adds a second orientation branch to each main-arm branch



$$2_{\text{shoulder}} \times 2_{\text{elbow}} \times 2_{\text{wrist}} = 8$$

- same feasible nonsingular tool pose
- distinct joint configurations
- joint limits and collisions can remove branches

How Many IK Solutions Can Exist?

The count depends on robot geometry, target location, and singularity

Representative regular cases

- planar 2R: usually two interior solutions
- spatial elbow 3R: up to four position solutions
- general spatial 6R: algebraic upper bound of sixteen

Where the count changes

- outside the workspace: no real solution
- workspace boundary: branches merge
- singular set: local structure changes
- constraints: mathematical branches may be infeasible

Polynomial roots are not automatically real, distinct, or feasible robot poses.

Turning Trigonometry into a Polynomial

The tangent-half-angle substitution exposes algebraic IK branches

$$a\sin\theta + b\cos\theta = c, \qquad u = \tan\frac{\theta}{2}$$

$$\sin\theta = \frac{2u}{1+u^2}, \qquad \cos\theta = \frac{1-u^2}{1+u^2}$$

After clearing the denominator

$$(b+c)u^2 - 2au - (b-c) = 0.$$

Real roots must be mapped back to angles and checked in the original equation. The projective case $\theta = \pi$ corresponds to $u = \infty$ and needs separate care.

A 6R Robot with Sixteen Real Solutions

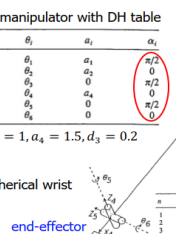
The familiar eight-branch count is not universal

an orthogonal manipulator with DH table

i	d_i	θ_i	a_i	α_i
1	0	θ_1	a_1	$\pi/2$
2	0	θ_2	a_2	0
3	d_3	θ_3	0	$\pi/2$
4	0	θ_4	a_4	0
5	0	θ_5	0	$\pi/2$
6	0	θ_6	0	0

$a_1 = 0.3, a_2 = 1, a_4 = 1.5, d_3 = 0.2$

with non-spherical wrist



Manseur and Doty:
International Journal of Robotics Research, 1989

solutions found using a fast numerical inversion algorithm ...

for the desired end-effector pose

$$ {}^0T_6 = \begin{bmatrix} -0.760117 & -0.641689 & 0.102262 & -1.140175 \\ 0.133333 & 0 & 0.991071 & 0 \\ -0.635959 & 0.766965 & 0.085558 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

there are 16 real solutions of the inverse kinematics!

all non-singular

n	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	$\det(J^2)$
1	0.000	107.458	112.460	−7.662	0.000	0.000	1.310
2	0.000	107.458	−67.540	−172.338	180.000	180.000	1.310
3	88.670	−176.682	−178.394	−63.284	157.829	139.944	−0.800
4	88.670	−176.682	1.606	−116.716	22.171	−40.056	−0.800
5	113.841	4.741	−179.093	−55.954	−63.659	−42.463	−1.256
6	113.841	4.741	0.907	−124.046	−116.341	137.537	−1.256
7	168.703	−104.205	146.556	−16.393	−170.903	98.216	0.803
8	168.703	−104.205	−33.444	−163.607	−9.097	−81.784	0.803
9	180.000	107.458	−147.375	−7.662	−164.675	180.000	0.732
10	180.000	107.458	32.625	−172.338	−15.325	0.000	0.732
11	−120.748	173.066	−178.472	31.328	−146.087	142.605	−0.717
12	−120.748	173.066	1.528	148.672	−33.913	−37.395	−0.717
13	−96.292	−5.766	−179.142	38.477	51.922	−39.631	−1.441
14	−96.292	−5.766	0.858	141.523	128.078	140.369	−1.441
15	−11.768	−105.495	−114.490	1.243	6.408	−79.398	1.318
16	−11.768	−105.495	65.510	178.757	173.592	100.602	1.318

Key counterexample

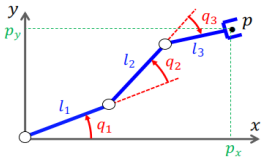
An orthogonal 6R robot without a spherical wrist can realize one desired pose with **16 distinct real nonsingular configurations**.

The eight-branch shoulder–elbow–wrist count is structure-specific.

Example adapted from De Luca, *Inverse Kinematics*.

Redundancy Creates Continua of Solutions

A redundant task can preserve the endpoint while internal joints move



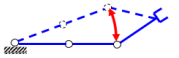
$l_1 = l_2 = l_3 = l \qquad n = 3, m = 2$

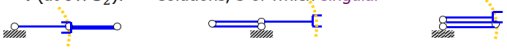
$WS_1 = \{p \in \mathbb{R}^2 : \|p\| \le 3l\} \subset \mathbb{R}^2$

$WS_2 = \{p \in \mathbb{R}^2 : \|p\| \le l\} \subset \mathbb{R}^2$

any planar orientation is feasible in WS_2

1. in $\mathrm{int}(WS_1) - \partial WS_2$: ∞^1 regular solutions at which the E-E can take a continuum of ∞ orientations (but not all orientations in the plane!)
2. if $\|p\| = 3l$ (at ∂WS_1): only 1 solution, singular
3. if $\|p\| = l$ (at ∂WS_2): ∞^1 solutions, 3 of which singular





4. if $\|p\| < l$ (in $\mathrm{int}(WS_2)$): ∞^1 regular solutions (that are never singular)

A planar 3R arm used only for point positioning has

$$n = 3, \qquad m = 2.$$

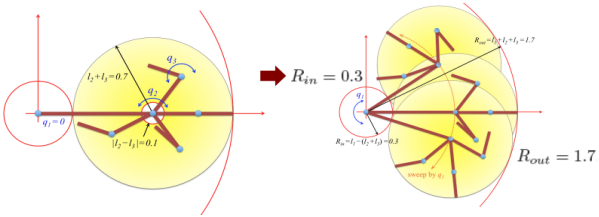
At a regular solution, the generic local solution-set dimension is

$$n - m = 1.$$

Thus one internal degree of freedom may vary while the point stays fixed.

Workspace of a Planar 3R Arm

The longest link decides whether a central hole remains



Sort the lengths as $l_{\min} \le l_{\text{med}} \le l_{\max}$.

$$R_{\text{out}} = l_1 + l_2 + l_3$$

$$R_{\text{in}} = \max\{0,\ l_{\max} - l_{\text{med}} - l_{\min}\}$$

$R_{\text{in}} > 0$ creates a central unreachable disk; otherwise the workspace includes the base.

Multiplicity by Dimension Count

Task dimension and Jacobian rank predict the local solution structure

Square task: $m = n$

Regular IK solutions are locally isolated and are usually finite in number.

Redundant task: $m < n$

Regular solutions generically form a local set of dimension

$$n - m.$$

A singularity changes the local structure

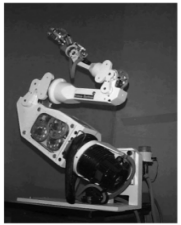
$$\operatorname{rank} J_r(q) < m, \qquad J_r(q) = \frac{\partial f_r(q)}{\partial q}.$$

Self-Motion of an 8R Arm

Redundancy changes internal posture while preserving the tool pose

- ∞^2 inverse kinematic solutions (redundancy degree $= n - m = 2$)





For a full spatial pose,
$n = 8, \quad m = 6, \quad n - m = 2.$
A two-dimensional family of joint motions can preserve the same tool pose at regular configurations.

Video checkpoint

Observe how Dexter changes its internal posture while preserving the tool pose.

Analytical or Numerical IK?

Choose the method by robot structure, constraints, and required output

Analytical methods

- fast evaluation after derivation
- expose discrete solution branches
- depend on exploitable geometry

Pieper-type conditions are sufficient geometric conditions for closed-form IK of many 6R arms.

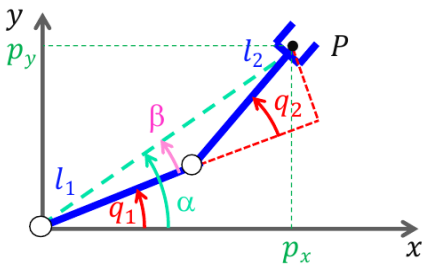
Numerical methods

- general across robot geometries
- natural for redundant tasks
- return a solution near the initial guess

Constraints, singularities, and convergence must be handled explicitly in either approach.

Planar 2R IK: Solve for the Elbow

One cosine equation gives reachability and both elbow branches



Square and add the two forward-kinematics equations:

$$c_2 = \frac{p_x^2 + p_y^2 - l_1^2 - l_2^2}{2 l_1 l_2}.$$

$$s_2 = \pm\sqrt{1 - c_2^2}, \qquad q_2 = \operatorname{atan2}(s_2, c_2).$$

Reachability test

$|c_2| > 1$ means the desired point lies outside the workspace.

Planar 2R IK: Solve for the Shoulder

Target bearing minus triangle angle completes each elbow branch

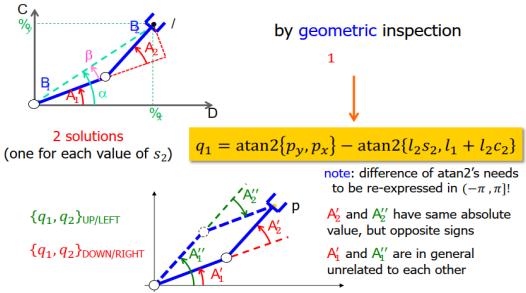
Define the target bearing

$$\alpha = \operatorname{atan2}(p_y, p_x).$$

Then, for each chosen sign of s_2,

$$q_1 = \operatorname{atan2}(p_y, p_x) - \operatorname{atan2}(l_2 s_2, l_1 + l_2 c_2).$$

Normalize the angle difference to the course convention.



Planar 2R IK: Algebraic Shoulder Solution

A linear system reproduces the geometric formula and reveals a singular exception

$$\begin{bmatrix} p_x \\ p_y \end{bmatrix} = \begin{bmatrix} l_1 + l_2 c_2 & -l_2 s_2 \\ l_2 s_2 & l_1 + l_2 c_2 \end{bmatrix} \begin{bmatrix} c_1 \\ s_1 \end{bmatrix}.$$

$$D = (l_1 + l_2 c_2)^2 + (l_2 s_2)^2 = l_1^2 + l_2^2 + 2 l_1 l_2 c_2.$$

For $D > 0$, solve for c_1, s_1 and recover

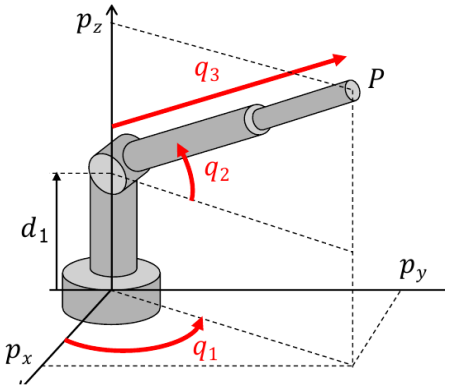
$$q_1 = \operatorname{atan2}(s_1, c_1).$$

Guard before division

If $l_1 = l_2$, $c_2 = -1$, and the target is the origin, then $D = 0$ and q_1 is undetermined.

Inverse Kinematics of a Polar RRP Arm

Radial extension, elevation, and azimuth can be solved in sequence



$$q_3 = \sqrt{p_x^2 + p_y^2 + (p_z - d_1)^2}$$

$$q_2 = \operatorname{atan2}\left(p_z - d_1, \sqrt{p_x^2 + p_y^2}\right)$$

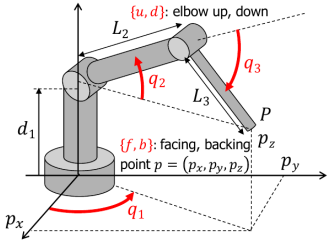
$$q_1 = \operatorname{atan2}(p_y, p_x)$$

Undefined-angle cases

Check $q_3 = 0$ and the axis case $p_x = p_y = 0$ before evaluating angles.

Geometry of a Spatial 3R Elbow Arm

Facing/backing and elbow up/down produce four regular position branches





For a symmetric arm without offsets:

- q_1: facing or backing branch
- q_3: elbow up or down branch
- q_2: completes each pair

$$2_{\text{base}} \times 2_{\text{elbow}} = 4$$

regular solutions in the interior of the workspace, away from the base axis.

Spatial 3R IK: Solve the Elbow Angle

Distance from the shoulder gives the same cosine-law structure as planar 2R

$$R^2 = p_x^2 + p_y^2 + (p_z - d_1)^2, \qquad c_3 = \frac{R^2 - L_2^2 - L_3^2}{2L_2L_3}.$$

Two elbow branches

$$s_3 = \pm\sqrt{1 - c_3^2}, \qquad q_3 = \operatorname{atan2}(s_3, c_3).$$

The positive and negative signs give the elbow-up and elbow-down configurations.

First gate

If $|c_3| > 1$, the desired point is outside the spherical-shell workspace.

Spatial 3R IK: Solve the Base Angle

The two base branches view the same radial line from opposite directions

When $p_x^2 + p_y^2 > 0$, define

$$q_1^{(f)} = \operatorname{atan2}(p_y, p_x), \qquad q_1^{(b)} = \operatorname{atan2}(-p_y, -p_x).$$

Facing branch

The signed radial coordinate

$$\rho = c_1 p_x + s_1 p_y$$

is positive.

Backing branch

The same expression for ρ is negative, changing the shoulder solution.

On the base axis, $p_x = p_y = 0$ and q_1 is undetermined: infinitely many base angles represent the same point.

Spatial 3R IK: Complete the Shoulder Angle

A 2 × 2 system determines q_2 for each base--elbow branch pair

Let

$$\rho = c_1 p_x + s_1 p_y, \qquad h = p_z - d_1.$$

Then

$$\begin{bmatrix} L_2 + L_3 c_3 & -L_3 s_3 \\ L_3 s_3 & L_2 + L_3 c_3 \end{bmatrix} \begin{bmatrix} c_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} \rho \\ h \end{bmatrix}.$$

Solve the system and recover

$$q_2 = \operatorname{atan2}(s_2, c_2).$$

Each choice of q_1 and q_3 must be propagated into this equation before q_2 is evaluated.

Branch Bookkeeping for the 3R Elbow Arm

A solution is a consistent tuple, not an independent choice for each joint

	elbow up $q_3^{(u)}$	elbow down $q_3^{(d)}$
facing $q_1^{(f)}$	solve matching q_2	solve matching q_2
backing $q_1^{(b)}$	solve matching q_2	solve matching q_2

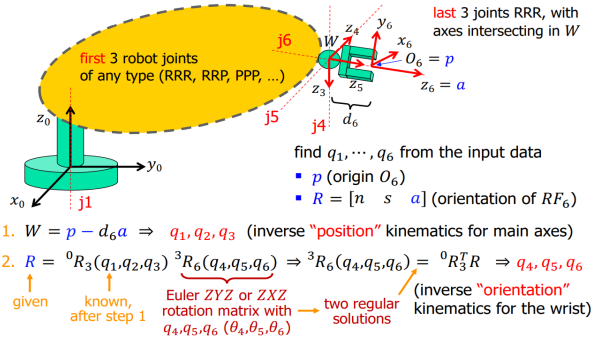
For each leaf

Substitute its q_1 and q_3 into the shoulder system, solve one matching q_2, then verify the complete tuple by forward kinematics.

Branches merge or become undefined on workspace boundaries and singular axes.

Spherical-Wrist Decoupling

Intersecting wrist axes separate position IK from orientation IK



The last three rotation axes intersect at the wrist center *W*.

$$W = p - d_6 a$$

where *p* is the tool origin and *a* is the tool *z* axis.

1. solve q_1, q_2, q_3 from *W*
2. solve q_4, q_5, q_6 from orientation

Position First, Orientation Second

The first-stage branch selects the rotation matrix used by the wrist stage

1. Main-arm position IK

$$W = p - d_6 a$$

Find every feasible tuple (q_1, q_2, q_3) placing the wrist center at W.

2. Wrist orientation IK

For each main-arm tuple, compute

$${}^3R_6 = ({}^0R_3)^T {}^0R_6.$$

Extract (q_4, q_5, q_6) as Euler angles, including both regular wrist branches.

Orientation solutions cannot be mixed across different main-arm branches.

Closed-Form IK of the PUMA 600

A spherical wrist makes the eight regular branches explicit

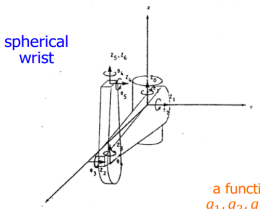


TABLE I
LINK PARAMETERS FOR PUMA ARM

Joint	α°	θ°	d	a	Range
1	-90°	θ_1	0	0	θ_1: +/− 160°
2	0	θ_2	0	a_2	θ_2: +45 to −225°
3	90°	θ_3	d_3	a_3	θ_3: 225° to −45°
4	-90°	θ_4	d_4	0	θ_4: +/− 170°
5	90°	θ_5	0	0	θ_5: +/− 135°
6	0	θ_6	0	0	θ_6: +/− 170°

$a_2 = 17.000$ $a_3 = 0.75$
$d_3 = 4.937$ $d_4 = 17.000$

here $d_6 = 0$,
so that $O_6 = W$ directly

a function of q_1, q_2, q_3 only!

$$
\begin{aligned}
n_x &= C_1[C_{23}(C_4C_5C_6 - S_4S_6) - S_{23}S_5C_6] \\
&\quad - S_1[S_4C_5C_6 + C_4S_6] \\
n_y &= S_1[C_{23}(C_4C_5C_6 - S_4S_6) - S_{23}S_5C_6] \\
&\quad + C_1[S_4C_5C_6 + C_4S_6] \\
n_z &= -S_{23}(C_4C_5C_6 - S_4S_6) - C_{23}S_5C_6
\end{aligned}
\Big\} \; n = {}^0x_6(q)
$$

$$
\begin{aligned}
o_x &= C_1[-C_{23}(C_4C_5S_6 + S_4C_6) + S_{23}S_5S_6] \\
&\quad - S_1[-S_4C_5S_6 + C_4C_6] \\
o_y &= S_1[-C_{23}(C_4C_5S_6 + S_4C_6) + S_{23}S_5S_6] \\
&\quad + C_1[-S_4C_5S_6 + C_4C_6] \\
o_z &= S_{23}(C_4C_5S_6 + S_4C_6) + C_{23}S_5S_6
\end{aligned}
\Big\} \; s = {}^0y_6(q)
$$

$$
\begin{aligned}
a_x &= C_1(C_{23}C_4S_5 + S_{23}C_5) - S_1S_4S_5 \\
a_y &= S_1(C_{23}C_4S_5 + S_{23}C_5) + C_1S_4S_5 \\
a_z &= -S_{23}C_4S_5 + C_{23}C_5
\end{aligned}
\Big\} \; a = {}^0z_6(q)
$$

$$
\begin{aligned}
p_x &= C_1(d_4S_{23} + a_3C_{23} + a_2C_2) - S_1d_3 \\
p_y &= S_1(d_4S_{23} + a_3C_{23} + a_2C_2) + C_1d_3 \\
p_z &= -(-d_4C_{23} + a_3S_{23} + a_2S_2).
\end{aligned}
\Big\} \; p = O_6(q)
$$

8 different (regular) inverse solutions
that can be found in closed form

For this geometry, $d_6 = 0$ and the tool origin coincides with the wrist center.

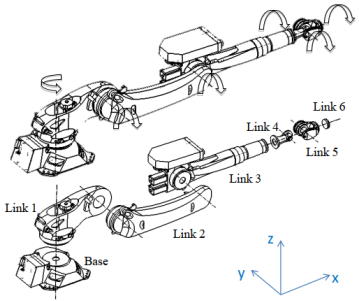
$2_{\text{shoulder}} \times 2_{\text{elbow}} \times 2_{\text{wrist}} = 8$ regular closed-form inverse solutions.

Joint limits, collisions, and singular poses still require filtering.

Forward Chain of a 6-DOF Industrial Robot

Separate the main arm, spherical wrist, and external frame transforms





$$\mathbf{T} = \mathbf{T}_{\text{Base}} \cdot \left[\overbrace{\mathbf{T}_z(d_0) \cdot \left\{ \mathbf{R}_z(q_1) \cdot \mathbf{T}_x(d_1) \cdot \mathbf{R}_y(q_2) \cdot \mathbf{T}_x(d_2) \cdot \mathbf{R}_y(q_3) \cdot \mathbf{T}_x(d_3) \cdot \mathbf{T}_z(d_4) \right\} \times}^{\text{Robot manipulator}} \underbrace{\left\{ \mathbf{R}_x(q_4) \cdot \mathbf{R}_y(q_5) \cdot \mathbf{R}_x(q_6) \right\} \cdot \mathbf{T}_x(d_6)}_{\text{Robot wrist}} \right] \cdot \mathbf{T}_{\text{Tool}}$$

$$T = T_{\text{Base}}\, T_{123}(q_1, q_2, q_3) \cdot T_{456}(q_4, q_5, q_6)\, T_{\text{Tool}}.$$

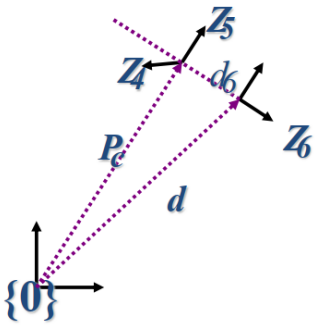
Structural split

- joints 1–3 place the wrist center
- joints 4–6 orient the tool
- base and tool transforms convert external conventions

Pieper's Step-by-Step Procedure

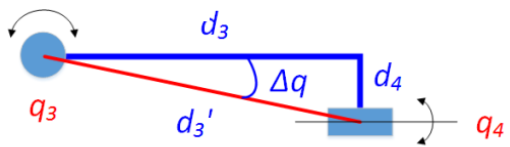
Use wrist-center geometry first, then solve the remaining rotation

1. express the desired tool pose as ${}^{0}T_{6}$
2. compute the wrist center $P_c = p - d_6 a$
3. set P_c equal to the position from ${}^{0}T_{3}(q_1, q_2, q_3)$
4. solve every main-arm branch (q_1, q_2, q_3)
5. compute ${}^{3}R_{6} = ({}^{0}R_{3})^{T 0}R_{6}$
6. solve every wrist branch (q_4, q_5, q_6)
7. preserve branch correspondence and filter constraints



Normalize Base, Tool, and Link Offsets

Convert the requested pose into the transform used by the internal kinematic model



$$d_3' = \sqrt{d_3^2 + d_4^2}, \qquad \Delta q = \operatorname{atan2}(d_4, d_3).$$

Remove known external transforms before solving joint variables:

$$T_0 = T_z^{-1}(d_0),\ T_{\text{Base}}^{-1},\ T,\ T_{\text{Tool}}^{-1},\ T_x^{-1}(d_6).$$

Absorb the perpendicular offset into the effective link length d_3' and angle shift Δq.

One normalized target keeps frame conventions out of the branch algebra.

Solve the Main Arm

After normalization, the position equations reduce to a planar 2-link problem

Let the normalized target position be (x', y', z') and define

$$\rho = \pm\sqrt{x'^2 + y'^2}.$$

Base branch

$$q_1 = \operatorname{atan2}(y', x')$$

for one radial sign; add the backing branch for the opposite sign.

Shoulder–elbow plane

$$\rho = d_2 \cos q_2 + d_3' \cos(q_2 + q_3 + \Delta q),$$

$$-z' = d_2 \sin q_2 + d_3' \sin(q_2 + q_3 + \Delta q).$$

Solve exactly as planar 2R IK.

If $x' = y' = 0$, the base angle is undetermined and must be handled as a singular case.

Solve the Robot Wrist

Each main-arm branch defines a residual rotation for the last three joints

$$T_{456} = T_{123}^{-1} T_0, \qquad R_{456} = R_x(q_4) R_y(q_5) R_x(q_6).$$

Regular wrist

Extract q_5 from a diagonal entry, then use paired off-diagonal entries in $\operatorname{atan2}$ to obtain q_4 and q_6.

Two regular Euler-angle branches represent the same residual rotation.

Wrist singularity

When $\sin q_5 = 0$, the first and third wrist axes align. Only a sum or difference of q_4 and q_6 is observable.

Choose one free angle explicitly and compute the other.

Numerical IK with Newton's Method

Linearize the task map at the current joint estimate

Let

$$e_k = r_d - f_r(q_k), \qquad J_k = J_r(q_k).$$

For a square nonsingular task, the Newton update is

$$q_{k+1} = q_k + J_k^{-1} e_k.$$

Why it can be fast

Near a regular solution, the local linear model is accurate and convergence can be quadratic.

Redundant task

For $m < n$, replace the inverse with a pseudoinverse:

$$\Delta q = J_k^{\dagger} e_k.$$

When Newton's Method Succeeds or Fails

The initial guess, conditioning, and step policy determine the observed branch

Reliable regime

- q_0 lies near a regular solution
- $J_r(q)$ is well conditioned
- the local linear step reduces task error

Common failure modes

- distant initial guess
- singular or nearly singular Jacobian
- step leaves joint limits or overshoots
- iteration converges to an unwanted branch

A numerical solver normally finds *one nearby solution*; multiple seeds are needed to explore multiple branches.

Gradient-Descent IK

Jacobian transpose gives a descent direction without matrix inversion

Define the squared task error

$$H(q) = \frac{1}{2} \|r_d - f_r(q)\|^2 .$$

Since

$$\nabla_q H(q) = -J_r^T(q) \left[r_d - f_r(q) \right] ,$$

steepest descent gives

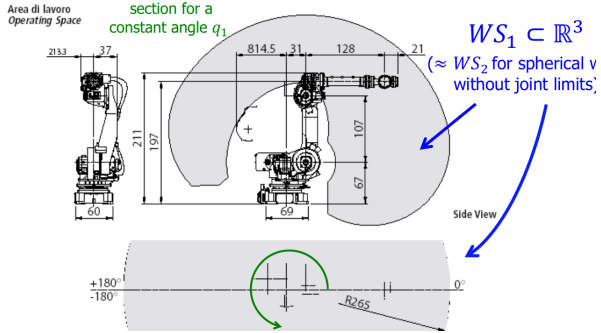
$$q_{k+1} = q_k + \alpha J_r^T(q_k) \left[r_d - f_r(q_k) \right] , \qquad \alpha > 0.$$

Step-size trade-off

Too large an α may increase the error or miss the minimum; too small an α makes convergence very slow.

Workspace of an Industrial Manipulator

A planar cross-section swept by the base joint creates a three-dimensional volume



Fix the base angle q_1 to inspect one meridional workspace section.

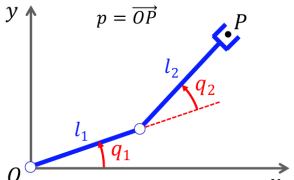
Rotating that section about the base axis sweeps the position workspace

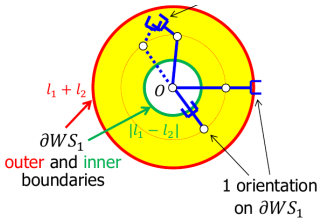
$$WS_p \subset \mathbb{R}^3.$$

With a spherical wrist and no joint limits, this approximates the wrist-center workspace.

Planar 2R Workspace and Branch Count

Reachability and IK multiplicity are two views of the same geometry





With $\rho = \sqrt{p_x^2 + p_y^2}$,

$$|l_1 - l_2| \le \rho \le l_1 + l_2.$$

- interior: two elbow branches
- inner or outer boundary: one merged branch
- outside: no real solution

If $l_1 = l_2$ and $\rho = 0$, infinitely many joint pairs fold to the origin.

Reference Materials

- A. De Luca — 09 DirectKinematics
- A. Klimchik — L1 Introduction
- A. De Luca — 10 InverseKinematics

Deserve “A” grade!

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Цифра, 521